

DARIUS FOO

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WORK EXPERIENCE

Research Fellow <i>National University of Singapore</i>	2026 – Present
Senior Software Engineer <i>SourceClear, later Veracode by acquisition</i>	2016 – 2020
Worked on the SourceClear Agent, a tool which uses a combination of static analysis and instrumentation to discover and automatically upgrade library dependencies from CI/CD pipelines.	

EDUCATION

National University of Singapore <i>PhD in Computer Science, under Graduate Tutorship-PhD Scheme</i>	2020 – 2026
- Thesis: <i>Automated Verification of Effectful Higher-order Programs</i> - Dean's Graduate Research Excellence Award, AY 2023/2024	
National University of Singapore <i>Bachelor of Computing (Honours) in Computer Science</i>	2012 – 2016

PROJECTS

Heifer an automated verifier for effectful higher-order programs
Choreographic PlusCal a choreographic specification language for TLA⁺
ppx_interact interactive breakpoints via the OCaml toplevel à la pdb/pry

SELECT PUBLICATIONS

Specification and Verification for Unrestricted Algebraic Effects and Handling Yahui Song, Darius Foo , Wei-Ngan Chin	ICFP 2024
Staged Specification Logic for Verifying Higher-Order Imperative Programs Darius Foo , Yahui Song, Wei-Ngan Chin	FM 2024
Protocol Conformance with Choreographic PlusCal Darius Foo , Andreea Costea, Wei-Ngan Chin	TASE 2023
Automated Temporal Verification for Higher-order Algebraic Effects Yahui Song, Darius Foo , Wei-Ngan Chin	APLAS 2022
Tracing OCaml Programs Darius Foo , Wei-Ngan Chin	OCaml 2022

ACADEMIC SERVICE

Programme Committee	OCaml 2025
Subreviewer	CAV 2022, POPL 2022, TASE 2024
Artifact Evaluation Committee	ESOP 2022
Student Volunteer	PLDI 2021, SPLASH 2022

TEACHING

Database Systems	2020, 2022, 2024 S1 & S2, 2025
Programming Methodology I & II	2013-2015, 2021-2023
Software Engineering	2016, 2022

SKILLS

OCaml, Lean, Rocq, JavaScript, Java; Spring Boot, React; PostgreSQL